

SMART INDIA HACKATHON 2026

Problem Statement ID — SIH26003

Problem Statement Title —

AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)

Theme — **MedTech / BioTech / HealthTech**

PS Category — **Software**

Organization — **Ministry of Development of North Eastern Region (MDoNER)**

Team Name (Registered on Portal) — **Design Divas**



SMRITI

(स्मृति / स्मृति)

AI-Powered Cognitive Care & Memory Companion for Dementia & MCI

- 5 Culturally Rooted Cognitive Games (NER Themes)
- Meta-WhatsApp Bot for Zero-Install Outreach
- Live-Render ML Adaptive Difficulty Engine
- 100% Offline-First PWA (Workbox + IndexedDB)
- Clinical Caregiver Command Center & Alerts

SOLUTION

Smriti is a hybrid cognitive-care ecosystem combining a WhatsApp AI companion bot, an offline-first PWA with 5 culturally-grounded cognitive games, and a clinical caregiver dashboard — built to meet elderly NER dementia patients where they already are (WhatsApp), instead of asking them to adopt new technology.

- **Patient Portal:** Frictionless PIN/biometric login, 5 cognitive games themed with real NER cultural context (Bihu Dhol, Kaji Nemu, Assam Tea, Bamboo Shoot).
- **Gemini AI Companion:** Voice-first, multilingual, grounded live in each patient's own medicine/schedule/family data.
- **WhatsApp Bot:** Daily reminders + auto-acknowledgment, zero app-install friction.
- **Caregiver Portal:** Real-time red-flag alerts, 7-day cognitive performance analytics, full patient management.
- **Live ML Engine:** Adaptive difficulty + cognitive health scoring live on Render.
- **True Offline-First PWA:** Full functionality with zero internet, auto-sync on reconnect.

PROTOTYPE

"English + Hindi live voice (Assamese text live, voice in progress), elder-friendly high-contrast UI, hands-free Web Speech API interaction, one-tap WhatsApp deep link."

**Patient PWA App**
56px touch targets, horizon routines, Family Photo Vault.

**WhatsApp Bot**
wa.me/15556680031 bridge, cron reminder dispatch.

**5 Cognitive Games**
Visual memory, sequencing, reminiscence, auditory rhythm.

**Caregiver Center**
Red-flag alerts, 1-tap call, 7-day adherence charts.

WHY WE STAND OUT ?

1. Real ML-Powered Engine
Live FastAPI model on Render calculates per-session difficulty and cognitive health scores from real gameplay telemetry, not static rules.

2. Culturally Authentic
Every game, name, and image grounded in real NER culture (Kaji Nemu, Bihu Dhol), unlike Western apps like Lumosity/BrainHQ.

3. True Offline-First PWA
Service Worker + IndexedDB caching works in zero-connectivity hill areas, verified via production PWA build.

4. Zero-Install WhatsApp
Patients unwilling/unable to install an app still get daily reminders and engagement through a platform they already use.

BACKEND ARCHITECTURE

- **REST & Auth:** Node.js/Express REST API, JWT + bcrypt auth (365-day persistent sessions, isolated Patient/Caregiver roles).
- **Database Layer:** MongoDB Atlas (Mongoose ODM): Patients, Caregivers, Reminders, GameSessions, Photos, Chat history.
- **Scheduled Cron:** node-cron scheduled reminder dispatch engine.
- **Conversational AI:** Google Gemini API powers both in-app chatbot and WhatsApp bot with live per-patient context injection.
- **ML Engine Microservice:** Teammate's independent FastAPI ML microservice (Render) exposing /get_next_difficulty and /calculate_health_score, called via mlService.js with a 6-second timeout + automatic rule-based fallback if unreachable.
- **Messaging:** Meta WhatsApp Business Cloud API: webhook receiver + wa.me deep link.
- **Offline Caching:** vite-plugin-pwa + Workbox + IndexedDB for offline-first caching and background sync queue.

THREE-TIER USER FLOW

- 1. Patient (Web/WhatsApp)**
PIN/biometric login or WhatsApp message → games/reminders/AI chat, synced to MongoDB.
- 2. Caregiver (Web Dashboard)**
Email/password or Google OAuth → scoped patient roster, red-flag alerts, cognitive analytics.
- 3. AI/ML Layer (Resilience)**
Gemini (conversational) + FastAPI ML service (adaptive difficulty & health scoring), each with graceful fallback.

AI COMPONENTS & INTEGRATION

Google Gemini LLM

Real-time conversational grounding using live patient data (medications, daily schedule, family relationships) with empathetic dementia care guardrails.

ML Difficulty/Scoring (Render)

Reaction time + mistakes + level → new difficulty + cognitive health score (0-100) + clinical status label (hosted at dementia-ai-engine.onrender.com).

Rule-Based Fallback Layer

Zero-latency accuracy-threshold logic backing every ML call, guaranteeing continuous uptime even during network drops.

FEASIBILITY & VIABILITY

✓ Technical Feasibility

Built entirely on free-tier, production-proven technology (React, Node.js, MongoDB Atlas, Gemini free tier, Meta WhatsApp Cloud API free tier) — no paid infrastructure required at pilot scale.

✓ Operational Feasibility

WhatsApp already has near-universal penetration across rural NER households; the app requires no new behavior from patients beyond messaging a known contact.

✓ Economic Feasibility

100% free and open-access to any rural clinic or ASHA center — no subscription, unlike CogniFit/Lumosity/BrainHQ (\$15–20/month, unaffordable for rural NER families).

✓ Regulatory Feasibility

DPDP Act 2023—aligned by design — explicit consent checkboxes, AES-256 encryption at rest, role-scoped access, named Grievance Officer, 30-day deletion guarantee.

CHALLENGES & MITIGATIONS

🔥 1. Zero digital literacy / no smartphone

Mitigation: WhatsApp-first design (no new app to learn); tiered fallback plan (SMS/IVR Tier 2, ASHA-assisted Kiosk Mode Tier 3) — architecture-ready, future scope.

🔥 2. Usability for cognitively-impaired elderly

Mitigation: 56px+ touch targets, zero hidden/hamburger navigation, high-contrast palette (following Apple Assistive Access / GrandPad patterns); voice-first interaction eliminates typing.

🔥 3. Reliability in NER's low-connectivity hills

Mitigation: True offline-first PWA (Service Worker + IndexedDB), verified via production build; background sync queue so no progress is lost when connectivity returns.

Economic Benefits

Reduces dependency on scarce, Guwahati-concentrated neurology specialists (85%+ of NER neurologists are based in Guwahati) by enabling remote, at-home cognitive engagement and caregiver-visible progress tracking — reducing costly, difficult specialist travel for families.

Social Benefits

Reduces elderly isolation through daily WhatsApp check-ins and culturally familiar engagement; empowers family caregivers with real visibility; helps counter regional stigma where cognitive decline is dismissed as normal aging ("বুঢ়া বয়সৰ পাহৰণি").

Systemic Benefits

Reduces unnecessary travel to distant specialist centers given NER's rugged terrain; supports the broader Digital India / DoNER telemedicine infrastructure push with structured longitudinal data.

STAKEHOLDERS & IMPACT FLOW

1. Patient

Receives a WhatsApp reminder or opens the PWA, plays a culturally familiar game or replies to a reminder.

2. Smriti AI Layer

Gemini interprets/responds conversationally; ML engine scores session and adjusts difficulty.

3. Caregiver

Sees real-time red-flag alerts and 7-day cognitive trend charts, intervenes only when needed.

4. Health / ASHA

Aggregated adherence and cognitive data informs community health worker rounds & regional monitoring.

OUR PROMISE & UN SDG ALIGNMENT

Aligns with **UN SDG 3 (Good Health & Well-Being)** and **SDG 10 (Reduced Inequalities — bridging the urban-vs-hill-state care gap)**. Directly supports MDoNER's own mandate of strengthening region-specific development and digital healthcare accessibility across the 8 NER states.

CLINICAL & EPIDEMIOLOGICAL DATA

- **LASI-DAD National Study (Lancet Public Health / PLOS ONE):** 8.8M Indian seniors with dementia (2026), 7.4-8.44% prevalence, projected 14.3M by 2036.
- **WHO / Alzheimer's Disease International:** ~55M global cases (2026) → 139M by 2050; 60%+ in LMICs.
- **NER-Specific Burden:** ~280,000-350,000 cases across 8 states; Assam alone ~180,000+; Vascular Dementia (driven by hypertension + high-sodium diets — Khar, Ngari, Shidol) is ~20% of cases, highest-in-NER subtype.
- **The "Neurology Desert":** 85%+ of NER's neurologists concentrated in Guwahati; hill states have near-zero full-time cognitive neurologists.

OUR WORKS & CODE REPOSITORIES

- **Live Frontend SPA:** Deployed on Vercel Engine
- **Live Backend API:** Node.js / MongoDB Atlas on Render
- **Live ML Microservice:** dementia-ai-engine.onrender.com
- **WhatsApp Companion Bot:** wa.me/15556680031
- **GitHub Repository:** github.com/gopalrajlohiya11-ui/Smriti

OPEN-SOURCE & ACADEMIC CITATIONS

- [LASI-DAD Diagnostic Study \(DOI\)](#) • [DementiaVoiceAnalyzer Repo](#)
- [NICE Guidelines CG42 \(CST\)](#) • ARDSI Guwahati/Mizoram • NIMHANS Tele-MANAS (14416)

WHY EXISTING SOLUTIONS FAIL HERE

- **CogniFit / Lumosity / BrainHQ:** \$15-20/month subscription (unaffordable for rural families), English-only, broadband-dependent — inaccessible to rural NER families.
- **MindMate / Timeless / CareZone:** Built for Western/NHS healthcare pathways, no Assamese/Hindi voice, no WhatsApp integration, break on 2G connections.
- **Smriti Solution:** 100% Free & open-access, culturally authentic NER themes, offline-first PWA, WhatsApp companion bot, and clinical ML telemetry.